

■ Retail Purchase Prediction Project

Problem

E-commerce platforms spend heavily on discounts. Not all users convert.

Goal: Predict buyers vs non-buyers to optimize promotions and reduce costs.

Approach

- Dataset: Online Shoppers Intention (12k+ sessions, 18 features).
- Preprocessing: encode categoricals, scale durations, stratify split.
- Model: Decision Tree baseline, Random Forest tuned with GridSearchCV.

Metrics

- Precision prioritized (targeting offers effectively).
- Recall, F1, ROC-AUC tracked.
- Best precision ~0.92, recall lower → accepted trade-off.

Insights

1. PageValues is the top predictor of buyer intent.
2. Random Forest reduces overfitting compared to single trees.
3. Business trade-off: focus on saving cost > capturing every buyer.